Xiangping Liu

Ph.D. Candidate in Biomedical Engineering | The University of Texas at Austin

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EDUCATION

The University of Texas at Austin, Cockrell School of Engineering — *Austin, TX, USA*
Ph.D. Candidate in Biomedical Engineering — *Aug 2022 - Present*

Jilin University, College of Chemistry — *Changchun, China*
B.Sc. in Chemistry — *Sep 2015 - Jun 2019*

Tsinghua University, Center for Life Science — *Beijing, China*
Summer School of Structural Biology, Chemical Biology and Medical Chemistry — *Summer 2017*

RESEARCH EXPERIENCE

Ultrasound-Induced CaO_2-Catalyzed Mechanoluminescent Nanoparticles for ChR2 Sono-Optogenetic Deep Brain Activation

Supervisor: *Prof. Huiliang (Evan) Wang*

- Developed Lipo@IR780/L012/CaO_2 liposomes with improved performance over first-generation organic blue mechanoluminescent nanoparticles by leveraging increased pH and H_2O_2 generation.
- Demonstrated sono-optogenetic neuromodulation of the ventral tegmental area (VTA).
- Evaluated neuronal activation after VTA stimulation using c-Fos staining.

Multicolored, Sonosensitizer-Optimized Organic Mechanoluminescent Nanoparticles for Functional Sono-Optogenetics

Supervisor: *Prof. Huiliang (Evan) Wang*

- Developed wavelength-tunable organic mechanoluminescent nanoparticles using FRET.
- Established a correlation between sonosensitizer energy gaps and ROS generation to guide sonosensitizer design and selection.
- Demonstrated bidirectional neuromodulation of genetically defined neurons using ChR2/ChRmine for activation and eOPN3 for inhibition.

PUBLICATIONS

† Co-first author. Xiangping Liu is shown in bold.

First- and Co-first-Author Publications

1. **Liu, X.**; Wang, W.; Artman, B.; Diao, J.; Zhao, Y.; He, W.; Yu, S.; Tang, K. W. K.; Yao, M.; Gu, C.; Song, B.; Wang, H. Multicolored, Sonosensitizer-Optimized Organic Mechanoluminescent Nanoparticles for Functional Sono-Optogenetics. *J. Am. Chem. Soc.* **2026**, 148, 16809-16820.
2. Gu, C.; **Liu, X.†**; Song, B.; Wang, W.; He, W.; Wang, H. Organic Mechanoluminescent Nanoparticles for Biomedical Applications. *Chem. Sci.* **2025**, 16 (28), 12702-12717.
3. **Liu, X.**; Hsieh, J.-C.; Markey, M.; Wang, H. iBMEntored Buddy Program for First-Year International BME Doctoral Students. *ASEE-GSW Proceedings* **2024**.
4. Wang, W.; Tang, K. W. K.; Pyatnitskiy, I.; **Liu, X.†**; Shi, X.; Huo, D.; Jeong, J.; Wynn, T.; Sangani, A.; Baker, A.; Hsieh, J.-C.; Lozano, A. R.; Artman, B.; Fenno, L.; Buch, V. P.; Wang, H. Ultrasound-Induced Cascade Amplification in a Mechanoluminescent Nanotransducer for Enhanced Sono-Optogenetic Deep Brain Stimulation. *ACS Nano* **2023**, 17 (24), 24936-24946.
5. Dang, Z.; **Liu, X.†**; Du, Y.; Wang, Y.; Zhou, D.; Zhang, Y.; Zhu, S. Ultra-Bright Heptamethine Dye Clusters Based on a Self-Adaptive Co-Assembly Strategy for NIR-IIb Biomedical Imaging. *Adv. Mater.* **2023**, 35 (46), e2306773.

Co-Author Publications

6. Wang, W.; Pyatnitskiy, I.; Shi, Y.; Tang, K. W. K.; Xie, Y.; Yu, S.; Wynn, T.; **Liu, X.**; Hsieh, J.-C.; Jeong, J.; He, W.; Artman, B.; Romero Lozano, A.; Shi, X.; Sangani, A.; Fenno, L. E.; Santacruz, S.; Chen, B.; Wang, H. Mechanoluminescent HOF Nanotransducers Enabled Sono-Optogenetics in Parkinsonian Rats. *Adv. Funct. Mater.* **2026**, e76189.

7. Wang, W.; Shi, Y.; Chai, W.; Tang, K. W. K.; Pyatnitskiy, I.; Xie, Y.; **Liu, X.**; He, W.; Jeong, J.; Hsieh, J.-C.; Lozano, A. R.; Artman, B.; Shi, X.; Hoefer, N.; Shrestha, B.; Stern, N. B.; Zhou, W.; McComb, D. W.; Porter, T.; Henkelman, G.; Chen, B.; Wang, H. H-Bonded Organic Frameworks as Ultrasound-Programmable Delivery Platform. *Nature* **2025**, 638 (8050), 401-410.

8. Tang, K. W. K.; Jeong, J.; Hsieh, J.-C.; Yao, M.; Ding, H.; Wang, W.; **Liu, X.**; Pyatnitskiy, I.; He, W.; Moscoso-Barrera, W. D.; Lozano, A. R.; Artman, B.; Huh, H.; Wilson, P. S.; Wang, H. Bioadhesive Hydrogel-Coupled and Miniaturized Ultrasound Transducer System for Long-Term, Wearable Neuromodulation. *Nat. Commun.* **2025**, 16 (1), 4940.

9. Narayan, O. P.; Dong, J.; Huang, M.; Chen, L.; Liu, L.; Nguyen, V.; Dozic, A. V.; **Liu, X.**; Wang, H.; Yin, Q.; Tang, X.; Guan, J. Reversible Light-Responsive Protein Hydrogel for on-Demand Cell Encapsulation and Release. *Acta Biomater.* **2025**, 193, 202-214.

10. Zhao, Y.-Y.; Zhang, Q.-W.; Liu, Y.-F.; Lv, C.; Guo, S.; **Liu, X.**-P.; Bi, Y.-G.; Li, H.-W.; Wu, Y.-Q. Improved Performance of $CsPbBr_3$ Quantum-Dot Light-Emitting Diodes by Bottom Interface Modification. *Org. Electron.* **2022**, 109, 106620.

11. Bai, L.; Hu, Z.; Han, T.; Wang, Y.; Xu, J.; Jiang, G.; Feng, X.; Sun, B.; **Liu, X.**; Tian, R.; Sun, H.; Zhang, S.; Chen, X.; Zhu, S. Super-Stable Cyanine@Albumin Fluorophore for Enhanced NIR-II Bioimaging. *Theranostics* **2022**, 12 (10), 4536-4547.

12. Han, T.; Wang, Y.; Xu, J.; Zhu, N.; Bai, L.; **Liu, X.**; Sun, B.; Yu, C.; Meng, Q.; Wang, J.; Su, Q.; Cai, Q.; Hettie, K. S.; Zhang, Y.; Zhu, S.; Yang, B. Surfactant-Chaperoned Donor-Acceptor-Donor NIR-II Dye Strategy Efficiently Circumvents Intermolecular Aggregation to Afford Enhanced Bioimaging Contrast. *Chem. Sci.* **2022**, 13 (44), 13201-13211.

13. Du, Y.; **Liu, X.**; Zhu, S. Near-Infrared-II Cyanine/Polymethine Dyes, Current State and Perspective. *Front. Chem.* **2021**, 9, 718709.

14. Liu, C.; Jin, Y.; Wang, R.; Han, T.; **Liu, X.**; Wang, B.; Huang, C.; Zhu, S.; Chen, J. Indole Carbonized Polymer Dots Boost Full-Color Emission by Regulating Surface State. *iScience* **2020**, 23 (10), 101546.

PRESENTATIONS

Multicolored Mechanoluminescent Nanoparticles for Functional Sono-Optogenetics

- **Oral Presentation**, 2025 MRS Fall Meeting & Exhibit, Boston, MA, Dec 2025.
- **Poster Presentation**, 23rd International Nanomedicine and Drug Delivery Symposium (NanoDDS 2025), Houston, TX, Oct 2025.

iBMEntored Buddy Program for First-Year International Doctoral Students

- **Poster Presentation**, BMES 2024 Annual Meeting, Baltimore, MD, Oct 2024.

TEACHING & MENTORING

CREATE Undergraduate Research Mentor *Summer 2026*

Mentoring a CREATE Summer Research Intern in HOF nanoparticle fabrication, focused ultrasound-triggered mechanoluminescence experiments, and nanoparticle characterization.

Undergraduate Research Mentor *Fall 2025 - Present*

Providing training in mechanoluminescent nanoparticle and HOF fabrication and related equipment use.

Undergraduate Research Mentor *Fall 2024 - Spring 2025*

Guided an undergraduate researcher in writing a review on organic mechanoluminescent nanoparticles for biomedical applications, published in Chemical Science.

iBMEntored Program - Mentor *Fall 2023 - Spring 2025*

Mentored first-year international BME doctoral students through the iBMEntored buddy program.

BME 365R Engineering Physiology I - Teaching Assistant *Fall 2023*

Led discussion sections and graded assignments. Instructors: Prof. Amy Brock and Prof. H. Grady Rylander III.

HONORS & AWARDS

2026 **Gregg and Marilyn Harris Endowed Graduate Fellow**
2026 **BME Professional Development Award**, The University of Texas at Austin
2025 **BME Professional Development Award**, The University of Texas at Austin
2022 **UT Austin Engineering Fellowship**
2019 **First Prize Scholarship**, Jilin University
2016 **Chinese National Endeavor Scholarship**

ACADEMIC SERVICE

Ad Hoc Reviewer - Nano Letters *2025 - Present*

Ad Hoc Reviewer - Biosensors and Bioelectronics *2026 - Present*

Ad Hoc Reviewer - Nature Biotechnology *2026 - Present*

SKILLS

- **Nanomaterials & Chemistry:** Organic synthesis; HOFs; polymeric nanoparticles; micelle/liposome formulation; drug loading; surface functionalization; bioconjugation; click chemistry.
- **Characterization & Imaging:** TEM; STEM; SEM; EDS; DLS; zeta potential; UV-Vis; fluorescence/chemiluminescence spectroscopy and imaging; NMR; HPLC; MALDI-TOF; confocal microscopy; NIR-II imaging.
- **Ultrasound & Neuroengineering:** Focused ultrasound (FUS); ultrasound-triggered drug delivery; sono-optogenetics; MEA recording; fiber photometry; laser speckle contrast imaging (LSCI).
- **Biological Techniques:** Mammalian cell culture; primary neuron preparation; stereotactic surgery; IV/intracranial injection; rodent neuromodulation; photothrombotic stroke models.
- **Software & Programming:** MATLAB; Python; C; ImageJ/Fiji; Origin; ChemDraw; Gaussian; MestReNova; Maestro; Spartan; EndNote; Cinema 4D; 3ds Max.